

A company intends to develop a budget management and approval system. The system aims to manage budgets proposed by departments, track the approval process, and manage budget allocations. The company uses five relational schemas described as follows:

1. DEPARTMENT(DeptID, Dept_Name, MngID)

The DEPARTMENT schema is used to store information about functional departments that propose budgets within the company. Each department has a unique department ID (DeptID), a department name (Dept_Name), and a manager identified by an employee ID (MngID).

2. EMPLOYEE(EmpID, Emp_Name, Emp_Role, DeptID)

The EMPLOYEE schema stores information about employees who are assigned as budget proposers or approvers. Each employee has a unique ID (EmpID), a name (Emp_Name), a position (Emp_Position), and belongs to a department (DeptID). The attribute Emp_Position can take one of the following values: “Proposer” or “Approver”.

3. BUDGET(ProposalID, Proposal_EmpID, Budget_Purpose, Amount, Status, Proposal_Date)

The BUDGET schema stores budget proposals submitted by employees. Each proposal has a unique ID (ProposalID), is submitted by an employee (Proposal_EmpID), and includes a purpose (Budget_Purpose). Each budget includes a proposed amount (Amount) measured in million VND, a status (Status), and a submission date (Proposal_Date).

The attribute Status can take the following values:

- “Proposed” – the budget has been submitted but not yet processed
- “Approved” – the budget has been approved
- “Rejected” – the budget has been rejected

4. APPROVAL(ApprovalID, ProposalID, Approval_EmpID, Approval_Date, Approved_Amount, Approval_Status)

The APPROVAL schema describes budget approval records. Each approval has an ID (ApprovalID) and may apply to one or more budget proposals. Therefore, the primary key consists of (ApprovalID, ProposalID). Each approval is performed by an employee (Approval_EmpID), on a specific date (Approval_Date), with an approved amount (Approved_Amount) measured in million VND for each proposal (ProposalID), and a status (Approval_Status) indicating “Approved” or “Rejected”.

Constraints:

- The approved amount (Approved_Amount) must not exceed the proposed amount (Amount).
- Each proposal (ProposalID) has at most one corresponding record in APPROVAL.
- If a proposal has status “Approved”, it must have a corresponding approval record with Approval_Status = “Approved”.
- If a proposal has status “Rejected”, it must have a corresponding approval record with Approval_Status = “Rejected”.

5. BUDGET_ALLOCATION(AllocationID, ProposalID, Allocation_Amount, Allocation_Date, Description)

The BUDGET_ALLOCATION schema stores information about budget allocation. Each allocation has an ID (AllocationID), is associated with a proposal (ProposalID), includes the

allocated amount (Allocation_Amount) in million VND, a allocation date (Allocation_Date), and a description (Description).

Constraints:

- Each approved proposal (Approval_Status = “Approved”) may have multiple allocations.
- The total allocated amount must not exceed the approved amount (Approved_Amount).

Question 1: Students are required to write the following queries using the specified language:

- List departments (DeptID, Dept_Name) where all budget proposals in 2024 have been approved (Status = “Approved”). **(SQL)**
- List budget proposals (ProposalID, Proposal_EmpID, Emp_Name) in 2024 that have been approved but whose total allocated amount is still less than the approved amount (Approved_Amount). **(SQL)**
- Find employees (EmpID, Emp_Name) with the position “Approver” who have not performed any approval in May 2025. **(Relational Algebra)**.
- For each department (DeptID, Dept_Name), count the number of proposals with amount less than 10 million VND that have been approved but have never received any budget allocation. **(Relational Algebra)**.
- Find all approvals (ApprovalID, Approval_Date, Approved_Amount) having proposer name (Emp_Name) “Trịnh Hoài An” and approver name “Phạm Tuấn Kiệt”. **(Relational Algebra)**.
- List departments (DeptID, Dept_Name) that have more than one employee with the position “Approver”. **(Relational calculus)**.
- f. Find employees (EmpID, Emp_Name) such that all of their budget proposals with amount ≥ 100 million VND have been approved. (Status = “Approved”). **(Relational calculus)**

Question 2: Specify all components (context, content, influence table/ integrity gate) of the following integrity constraints:

- If a budget has status “Approved”, it must have a corresponding approval record with status “Approved”.

If a budget has status “Rejected”, it must have a corresponding approval record with status “Rejected”.

- b. Each approval ID (ApprovalID), possibly associated with multiple proposals, must be performed by the same employee (Approval_EmpID).

Question 3: Consider the relation Assignment with functional dependencies F:

ASSIGNMENT(InstructorID, CourseID, Role, StartDate, EndDate, TeachingHours, Credits, Phone, Email)

Primary key: {InstructorID, CourseID, Role}

F = {

- f1: InstructorID $\rightarrow$ Phone, Email
- f2: CourseID, Role $\rightarrow$ TeachingHours
- f3: CourseID $\rightarrow$ Credits

}

- a. Give an example of attribute redundancy in the *ASSIGNMENT* relation.
- b. Determine the normal form of the *ASSIGNMENT* relation. Explain your answer.
- c. Normalize the *ASSIGNMENT* relation to achieve at least Third Normal Form (3NF). Explain your steps.